GH RAISONI COLLEGE • DEPARTMENT OF BBA-CA • PUNE CAMPUS

# AI CHATBOT FOR COLLEGE — REVISED SYSTEM SPECIFICATIONS

*Explicit Database Swimlane showing where student records are stored and fetched*

## 1. Activity Diagram (With Dedicated Database Swimlane)

✔ Faculty Action Item: Explicit Database Representation in Activity Flow

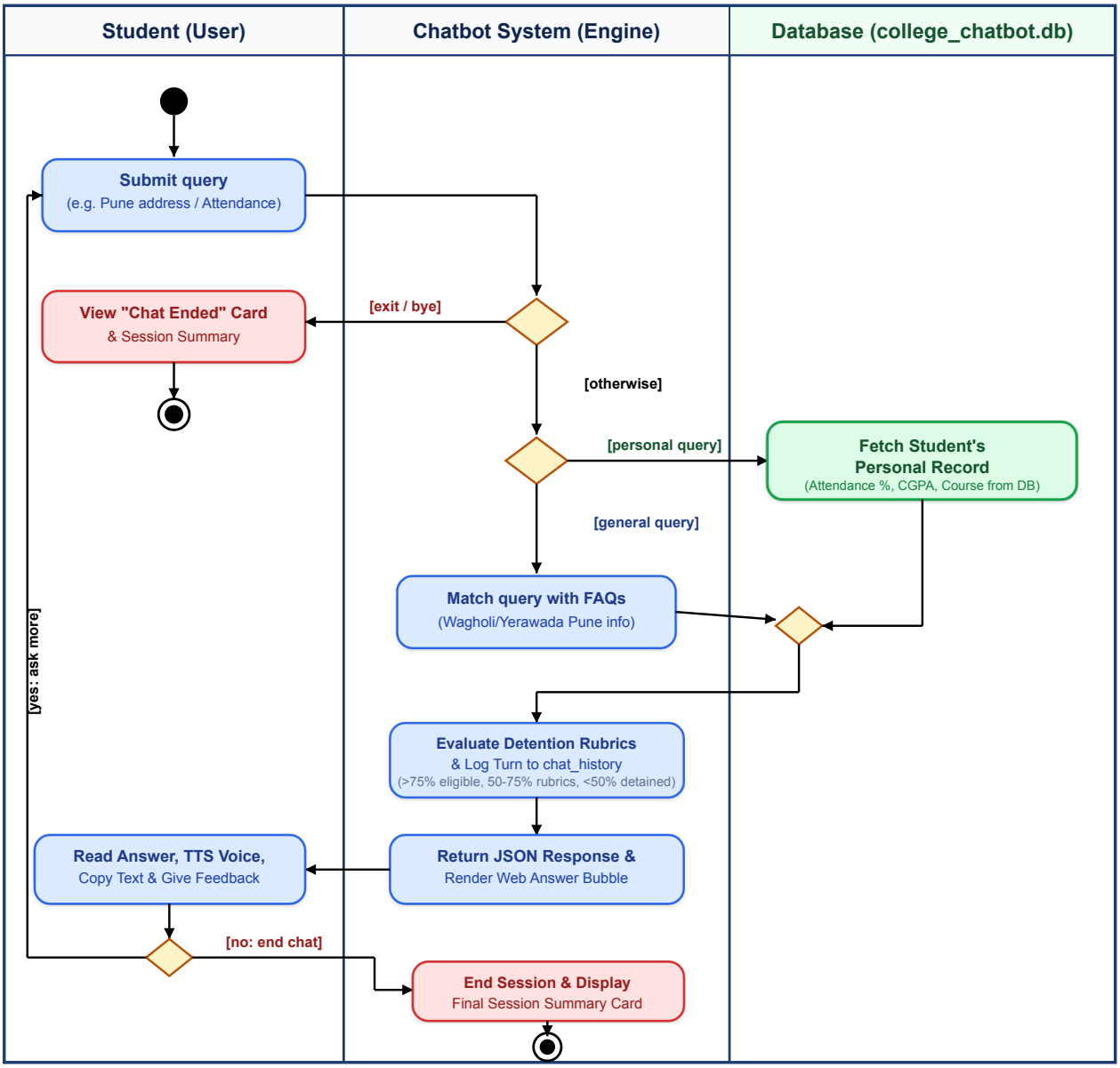



*Fig 1: Activity Diagram — 3-Swimlane architecture (Student, Chatbot System, Database) showing explicit personal record fetching and chat logging.*

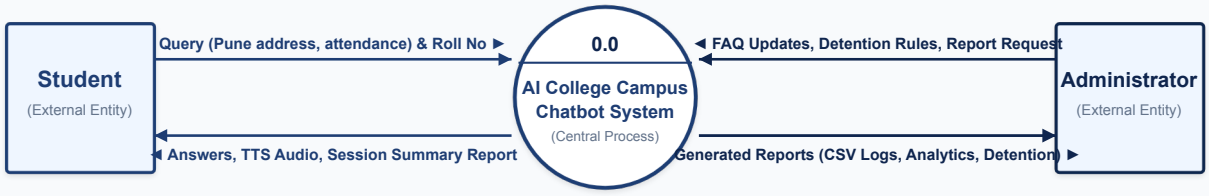
# AI CHATBOT FOR COLLEGE — REVISED SYSTEM SPECIFICATIONS

*First-Level data flow decomposition with external entities, processes, data stores, and symbol guide*

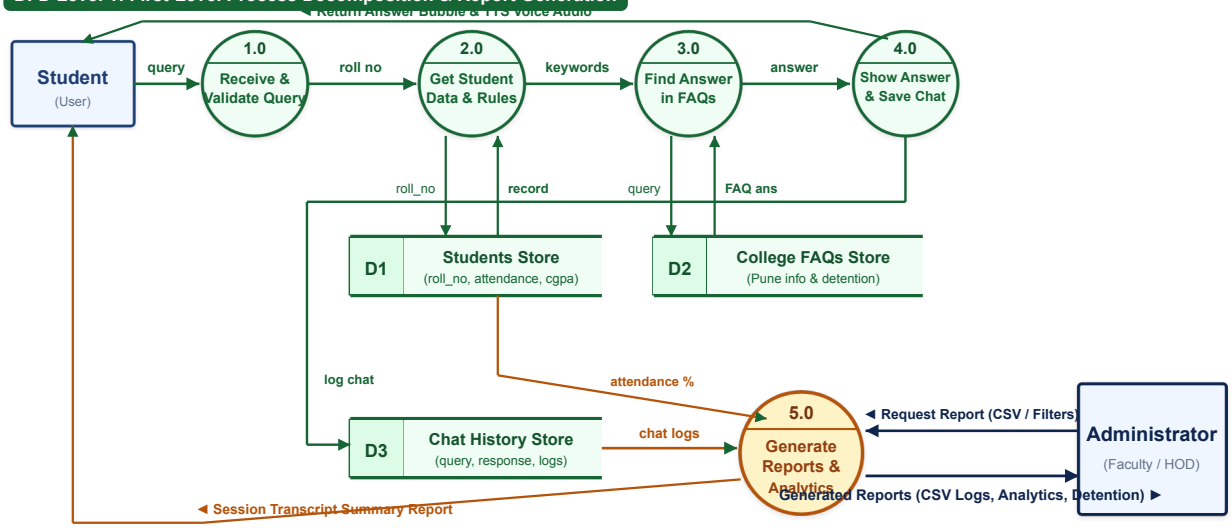
## 2. Data Flow Diagram (DFD Level 0 Context & Level 1 Process Breakdown with Report Generation)

✔ Faculty Action Item: First-Level DFD & Complete Symbols Table

**DFD Level 0: Context Diagram**




**DFD Level 1: First-Level Process Decomposition & Report Generation**




**Standard DFD Notation & Symbol Conventions (Gane-Sarson / DeMarco)**

| Symbol Shape | DFD Element | Standard Academic Definition | Implementation in Our Project |
|---|---|---|---|
|  | External Entity | Origin/destination of data outside boundary (Square / Rectangle) | Student (User) & Administrator (Faculty/HOD) |
|  | Process (Bubble) | Transforms incoming data into outgoing data (Circle with ID) | 1.0 to 4.0 Query Handlers & 5.0 Generate Reports |
|  | Data Store (Open) | Repository of data at rest (Open-ended parallel lines; NEVER closed) | D1 Students, D2 College FAQs, D3 Chat History |

*Fig 2: Data Flow Diagram (DFD) — Properly formatted Level 0 Context Diagram (System boundary with Student & Administrator) plus Level 1 Process Decomposition (1.0 to 4.0 Query Handlers & 5.0 Generate Reports & Analytics) with standard open-ended data stores (D1 Students, D2 FAQs, D3 Chat History) and formal academic DFD symbols reference table.*

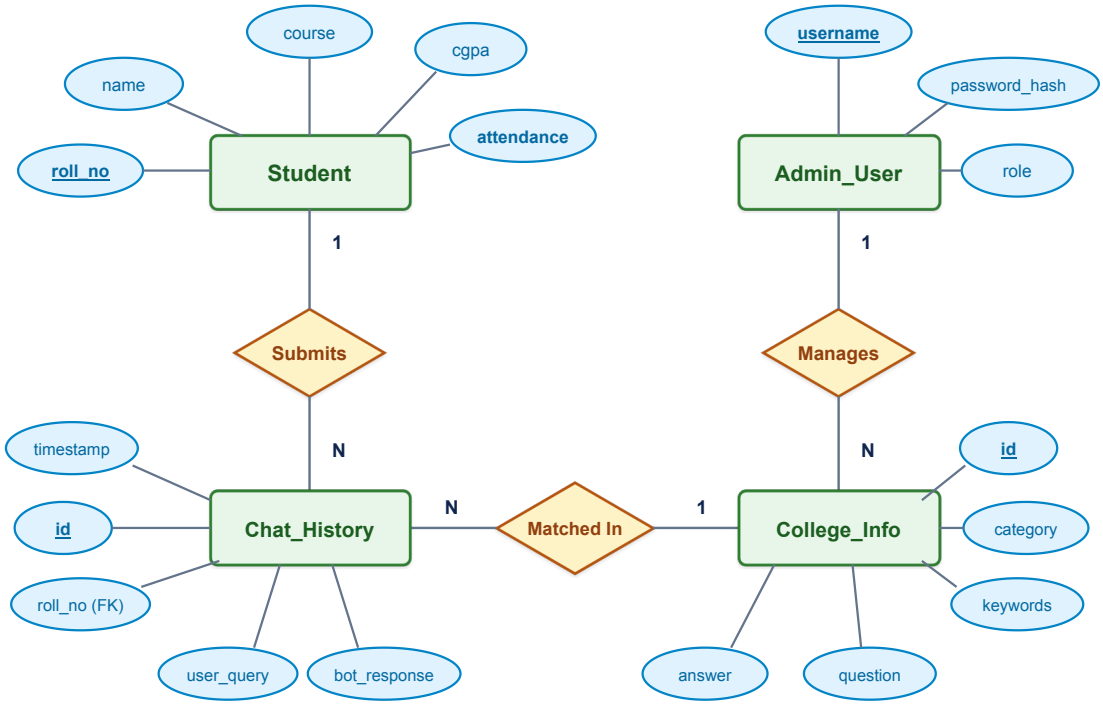
# AI CHATBOT FOR COLLEGE — REVISED SYSTEM SPECIFICATIONS

*Classic ER modeling with 4 core entities, attributes, cardinalities, and symbol legend*

## 3. Entity-Relationship (ER) Diagram — Classic Notation

✔ Faculty Action Item: Classic ER Notation matching Faculty Specification

### ER Diagram (Classic Notation): AI Chatbot for College




**Schema & Project Knowledge Base Defense:**

- **Student Entity:** Stores attendance %, CGPA & evaluates 4-tier detention policy.
- **College_Info Entity:** Contains Pune campuses (Wagholi & Yerawada) & exam rules.
- **Chat_History Entity:** Logs each turn with student roll_no FK, query & bot answer.
- **Admin_User Entity:** RBAC authentication for updating college FAQs & detention data.

*Verified in 3NF Normalization: Zero transitive dependencies & lossless join preservation.*

**Key / Legend:**

- Entity (Table)
- Attribute (Column) — id = Primary Key
- Relationship
- Link / Cardinality (1:N)

*Fig 3: Entity-Relationship Diagram (ERD) — Classic notation with Student, Admin_User, Chat_History, and College_Info.*

# AI CHATBOT FOR COLLEGE — REVISED SYSTEM SPECIFICATIONS

*Detailed structural justification, attribute breakdown, normalization, and relational flow*

## 4. Entity Review & Justification (3NF Schema Analysis)

✔ Faculty Action Item: Formal Entity Review & Relational Justification

### ENTITY REVIEW & JUSTIFICATION MANIFESTO (3NF SCHEMA ARCHITECTURE)

Comprehensive analysis of the 4 core entities, structure, attributes, normalization, and relational flow as requested in faculty action items.

| Entity Name | Primary Key | Foreign Key(s) | Key Attributes | System Role & Justification |
|---|---|---|---|---|
| **Student** | `roll_no` | None (Base Entity) | name, course, cgpa, attendance | Represents enrolled students. Enables personalized query responses regarding attendance percentage, detention risk, and academic standing. |
| **Admin_User** | `username` | None (Independent) | password_hash, role | Stores administrative credentials with salted SHA-256 hashes. Restricts knowledgebase CRUD operations and audit log exports to authorized staff. |
| **College_Info** | `id` | None | category, keywords, question, answer | The institutional knowledge base. Contains categorized FAQ entries (Admissions, Fees, Exams, Pune Campus Location, Detention Rubrics) matched by the NLP engine. |
| **Chat_History** | `id` | `roll_no` (FK) | user_query, bot_response, timestamp | Audit telemetry and analytics log. Records every student interaction, resolved FAQ answer, timestamp, and feedback for administrative review and training. |

**Relational Cardinality & Integrity Rules:**

- **Student (1) — Submits — (N) Chat_History:** A registered student can submit multiple inquiries over time. If a query is submitted anonymously by a guest, roll_no is set to NULL (optional 1:N).
- **College_Info (1) — Matched In — (N) Chat_History:** A specific FAQ record can match multiple historical student questions across sessions.
- **Admin_User (1) — Manages — (N) College_Info:** Authorized department administrators create and update knowledgebase records.

**Why Exactly These 4 Entities? (Normalization Defense)**

- **Eliminates Redundancy:** Separating student personal data from chatbot logs prevents duplicating student names/CGPA in every chat record.
- **Third Normal Form (3NF):** Every non-key attribute depends strictly on the primary key, the whole key, and nothing but the key.
- **Security Isolation:** Administrator credentials remain physically and logically isolated from public FAQ and student interaction data.

*Fig 4: Entity Review & Justification Table — Formal 3NF relational breakdown of the 4 core database entities.*

# AI CHATBOT FOR COLLEGE — REVISED SYSTEM SPECIFICATIONS

*Official Theory and Practical Detention Rubrics trained into chatbot knowledge base*

## 5. Implementation of Student Detention Policy

✔ Faculty Action Item: Official Detention Policy Integration & Rubrics

### OFFICIAL STUDENT DETENTION POLICY & CHATBOT LOGIC IMPLEMENTATION

Trained into Chatbot NLP Knowledge Base and Personalized Response Logic as clarified in Faculty Meeting.

### 1. THEORY DETENTION RUBRICS

(Attendance Excluding Medical Grounds)

| S.No | Attendance Bracket | Academic Rubrics & Requirements | Eligibility Status |
|---|---|---|---|
| — | Above 75.0% | Regular academic standing. No additional rubrics required. | ✔ Fully Eligible |
| 1 | 60.0% to 75.0% | End Sem Examination Paper Sets of previous 2 years to be solved in **Open Book mode** in the College. | Eligible after solving sets |
| 2 | 50.0% to 59.99% | (1) End Sem Examination Paper Sets of previous 2 years solved in Open Book mode in College.<br>(2) Must appear for a prelim examination and score a **minimum of 60% marks** to get out of detention. | Eligible if prelim ≥ 60% |
| 3 | Below 50.0% | **No consideration. Strictly detained from university end-semester examinations.** | ✘ Strictly Detained |

### 2. PRACTICAL DETENTION RUBRICS

(Note: Not more than 2 practicals to be conducted in a day)

| S.No | Attendance Bracket | Practical Rubrics & Penalties | Practical Eligibility |
|---|---|---|---|
| 1 | 60.0% to 75.0% | The student will **not be eligible for the attendance mark component** of internal practical marks. | Attendance marks forfeited |
| 2 | 50.0% to 59.99% | Student must appear for a **practical-based prelim examination** and score a minimum of 60% marks to be eligible to get out of detention. | Eligible if practical ≥ 60% |
| 3 | Below 50.0% | **No consideration. Strictly detained from practical examinations.** | ✘ Strictly Detained |

**Chatbot Engine Integration Details:**

When a logged-in student asks *"What is my attendance?"*, app.py evaluates their percentage against these exact 4 thresholds and returns their specific status, eligible exam status, and designated academic rubrics. Both Pune Campus addresses — Wagholi Campus (Gat 1200, Domkhel Rd, Pune - 412207) and Yerawada Campus (Nyati Unitree, Airport Rd, Yerawada, Pune - 411006) — are active in institutional FAQs.

*Fig 5: Implementation of Student Detention Policy — Official Theory and Practical rubrics table trained into chatbot engine.*